01 — TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

PAGE / 08

Two-Color Flip Counters, Solid '10' Rods & Dual 10-Frame Lid

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? PUT AWAY YOUR FINGERS & SEE NUMBERS AT A GLANCE.

Your brain can instantly recognize up to 5 objects without counting. This 2×5 Ten-Frame turns addition and "carrying the 1" into visual patterns!

5-ANCHOR SUBITIZING & TWO-COLOR COUNTERS

Recognizing a small quantity without counting one-by-one is called "subitizing." Our Ten-Frame organizes numbers into two rows of 5:

- **Two-Color Counters: Flip between Red (Side A) and Green (Side B)**
- **Tactile Finger Rings: Grooved surfaces you can feel without looking**
- **Stackable Plugs: 10 counters snap together into a solid 10-tower**

Once you see 8 as "a full row of 5 plus 3 more," finger-counting disappears!

10 ONES = 1 SOLID TEN-ROD (70 MM)

Stack 10 unit counters into a tower: it stands 70.0 mm tall — the exact same height as one solid purple '10' rod! That's place value in your hand.

WHAT'S INSIDE YOUR BOX (54 PIECES)

- 40 Two-Sided Reversible Counters (Red on Side A / Green on Side B)
- 6× Solid '10' Regrouping Rods + 6× Operator Blocks (+, −, =, 10) + Box & Lid
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

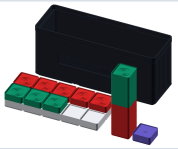
1. **FLIP THE TWO-COLOR COUNTERS:** Notice every counter is Red on one side and Green on the other—flip to show two addends!
2. **FILL ONE ROW OF 5 ON THE LID:** Drop 5 Red counters along the top row and 3 Green below: your eyes see 5 + 3 = 8 instantly!
3. **PRESS THE BOX ICON KEY [▦]:** Type 8 [ENTER] 2 [÷] 5 [÷] [▦] on your calculator—the 2×5 grid icon on your box is the [▦] key!

BOX ICON = CALCULATOR [▦] TEN-FRAME & [PV] PLACE VALUE KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([▦] Ten-Frame & [PV])! Build every lesson with your blocks, then press the key.

5-ANCHOR SUBITIZING (5+3=8) ↔ BOX [▦] & [PV] ICONS

STEP A: HANDS-ON WITH 5-ANCHOR SUBITIZING — 1:1 CAD SCENE



Top Row = 5 Instant Benchmark | 5 Red + 3 Green See 8 at a Glance | 2 Empty Sockets 10 − 8 = 2

▲ Because each 10-frame is arranged in two rows of 5, your eyes recognize 5 + 3 = 8 and 2 empty spots instantly!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

5-ANCHOR TREE: 8 = 5 + 3 (2 SHY OF 10)

10-FRAME CAPACITY = 10

Filled: 5+3 = 8 | 10 − 8 = 2 [PV]

Top Row: 5 Red | Bot Row: 3 Green | Empty = 2

5 [ENTER] 3 [+] 10 [xey] [−] [PV] → 2

PRESS BOX ICON & RPN KEYS (SC6):

1. 5-ANCHOR: 5 ENTER 3 + = 8

2. 10-COMP: 10 xey − PV = 2

✓ DISPLAY (X) = 2

KEYSTROKES: 5 [ENTER] 3 [+] 10 [xey] [−] [PV]

Build 5 + 3 = 8 on the lid tray, then subtract from 10 and press [PV] to verify the 2 empty sockets!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

02 — HOW IT WORKS: THE CORE IDEA

PAGE / 08

5-Anchor Subitizing & Trading 10 Ones for 1 Ten-Rod

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

WHY WE GROUP IN TENS (PLACE VALUE)

Our entire number system is built on powers of 10:

10 × 10^k = 10^(k+1)

Whenever ten ones fill up a Ten-Frame, they trade up into a single unit of the next higher place value:

- **10 Unit Counters (10 × 1) = 1 Solid Ten-Rod (1 × 10)**
- **10 Ten-Rods (10 × 10) = 1 Hundred-Flat (1 × 100)**

Because the Ten-Frame physically holds at most 10 counters, "carrying" and "borrowing" stop being mysterious rules — you see the tray fill up!

HARMONIZED 7.00 MM STACKING HEIGHT

Each counter adds 7.00 mm of height (1U), matching our 1/12th Fraction block and Prime block [2]. A stack of 10 counters is 70.0 mm tall — matching the solid '10' rod right down to the millimeter.

TACTILE GROOVES & EASY LIFT NOTCHES

Every counter has concentric finger rings you can feel with your eyes closed, and every tray socket has side thumb-notches so small fingers can pop counters out effortlessly.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ **TRAP 1 — Counting on fingers one-by-one:**
✓ **FIX:** The 2×5 grid trains your brain to see 8 as '5 and 3' or '2 shy of 10' in a split second.

✗ **TRAP 2 — Blindly 'carrying the 1' without meaning:**
✓ **FIX:** Trading 10 stacked counters (70mm) for 1 Purple '10' rod (70mm) proves what that '1' actually is!

✗ **TRAP 3 — Forgetting 10-complements (7 + ? = 10):**
✓ **FIX:** Any partially filled 10-frame shows the complement as the empty sockets staring right back at you.

CALCULATOR CHECK: 5 [ENTER] 3 [+] 10 [xey] [−] [PV] → X = 2

COMPONENT & DIMENSION GUIDE

COMPONENT	DIMENSIONS	VALUE	COLOR
Dual Counter	Ø18.0 × 7.00 mm	1 Unit	Red / Green
10-Stack Tower	Ø18.0 × 70.0 mm	10 Units	Red / Green
Solid '10' Rod	19.6 × 70.0 mm	1 Ten	Purple
Operator Block	19.6 × 19.6 mm	+ − =	Gold
Ten-Frame Lid	126 × 42 × 10 mm	2x5 Tray	Matte Black
3x1 Storage Bin	126 × 42 × 49 mm	4 Towers	Matte Black

WHY YOUR BRAIN LOVES THIS

- Instant Visual Benchmarks: A full row is always 5; a full tray is always 10.
- Easy Finger-Lift Notches: Counters pop in and out without frustration.
- Stackable Into Towers: Snap 10 counters together to build a 10-rod.
- Fast Cleanup: 4 wells hold 10 counters each — if all 4 are level, none are lost!

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85mm

7mm — 1U = 7.00 mm VISIBLE DATUM

BASE PLUG (1.85mm) ▼ 1.85mm

✓ FLUSH PLUG-AND-SOCKET JOINT

Socket swallows plug completely: Only 7.00mm true heights add!

Box Button Icon Keys: [▦] Ten-Frame & [PV] Place Value

5-Anchor Subitizing: 8 = 5 + 3 | 7 = 5 + 2 | 9 = 5 + 4

10-Complement Law: n + (10 − n) = 10 (Filled + Empty = 10)

Make-a-Ten Addition: 8 + 7 = (8 + 2) + 5 = 10 + 5 = 15

Tens & Ones Keystroke: N [ENTER] 10 [g][÷R] → X Tens, Y Ones

BASE-10 PLACE VALUE • 10 UNIT COUNTERS = 1 SOLID TEN-ROD

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

03 — EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Instant Recognition: See 5 Red + 3 Green = 8 Without Counting

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

1 SUBITIZING 8 & BRIDGING TEN (8 + 7 = 15)

1. **MODEL 5-ANCHOR SUBITIZING (NUMBER 8):**
   • Place 5 Red counters in the top row and 3 Red in the bottom row.
   → Eyes see 5 + 3 = 8 and 2 empty sockets (8 + 2 = 10)!
2. **BRIDGE ACROSS 10 TO ADD 8 + 7:**
   • Start with 8 Red counters in the frame and 7 Green counters outside.
   • Slide 2 Green counters into the 2 empty sockets to complete 10!
3. **READ THE MAKE-A-TEN SUM AT A GLANCE:**
   1 full Ten-Frame (10) + 5 Green counters remaining outside:
   → 8 + 7 = (8 + 2) + 5 = 10 + 5 = 15!

BENCHMARK 5 AND 10 MENTAL ANCHORS

Cognitive research proves that students who master 5 and 10 benchmarks calculate 3x faster than those relying on counting by ones. The physical ten-frame hard-wires these geometric benchmarks into working memory.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [▦] Ten-Frame & [P

STEP 1 (FILL TO 10): Slide 2 Green counters into the left 10-frame on the lid and press 8 [ENTER] 2 [÷] (= 10).

STEP 2 (ADD REMAINING 5): Press 5 [÷] → X = 15 (matching 1 full 10-frame + 5 Green counters in the right frame).

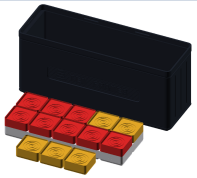
STEP 3 (PRESS BOX ICON [▦]): Press [▦] on your SC-6 calculator to display 15 as a live visual Ten-Frame on screen!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):

T: 0 | Z: 0 | Y: 0 | X: 15 → DISPLAY (X) = 15

LESSON 1: MAKE-A-TEN (8+7 = 10+5) ↔ [▦] TEN-FRAME KEY

STEP A: HANDS-ON WITH COUNTERS ON LID — 1:1 CAD SCENE



Left 10-Frame 8 Red + 2 Green = 10 | Right 10-Frame 5 Green Remaining | Total Sum 10 + 5 = 15

▲ Slide 2 Green counters across to finish filling the left 10-frame: 8 + 7 transforms into 10 + 5 = 15!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

MAKE-A-TEN NUMBER BOND: 8 + 7 = 15

8 + 7 = 15 [▦]

8 + 2 = 10 (Full) | 7 Splits: 2 + 5

8 Red | +2 Green | +5 Green

8 [ENTER] 2 [÷] 5 [÷] [▦] → 15

PRESS BOX ICON & RPN KEYS (SC6):

1. MAKE 10: 8 ENTER 2 + = 10

2. BOX ICON KEY: 5 + ▦ = 15

✓ DISPLAY (X) = 15

KEYSTROKES: 8 [ENTER] 2 [÷] 5 [÷] [▦]

Slide 2 counters on the lid to make 10, then press the Box Icon key [▦] on your SC-6 calculator!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 15 → X = 15

SUBITIZING SUPERPOWER • SEE NUMBERS IN CHUNKS OF 5 AND 10

CONTINUE RIGHT → PAGE 04

04 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Make-a-Ten Bridge: Why 8 + 7 Regroups Into 10 + 5 = 15

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

2 MAKING TEN REGROUPING: 8 + 7 = 15

THE PROBLEM: Add 8 + 7 without counting on fingers.

1. FILL FRAME 1 WITH 8 RED COUNTERS:
   Top row has 5; bottom row has 3. Exactly 2 spaces are empty.
2. DECOMPOSE NUMBER 7 INTO 2 + 5:
   • Take 2 green counters from the 7 and move them to Frame 1.
   → Frame 1 is now FULL (8 + 2 = 10)!
3. REGROUP INTO PLACE VALUE BLOCKS:
   • Swap the full Frame 1 for one solid '10' miniblock rod.
   • The remaining 5 green counters sit in Frame 2.
   → Total: 1 Ten + 5 Ones = 15!

✓ THE "MAKING TEN" BRIDGE THEOREM

8 + 7 = (8 + 2) + 5 = 10 + 5 = 15

Carrying in columns is simply filling a physical frame.

Students develop permanent mental addition agility by seeing the physical conservation of counters across two frames.

★ 2-PLAYER GAME: 'FLASH TEN-FRAME SNAP!' (2 MINS)

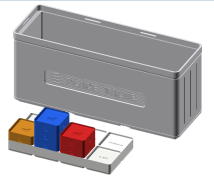
Player A secretly arranges counters on the lid's ten-frame behind a book, lifts the book for just 1 second, and covers it again. Player B must call out both the number of counters AND **how many empty spots are needed to make 10, then check with [▦]!**

⚙ WHERE THIS SHOWS UP IN COMPUTER MEMORY & DIGITAL LOGIC

Grouping individual units into fixed-size frames (10 in decimal, 8 bits = 1 byte in binary) is the foundation of positional number systems! When a frame fills up, it overflows ('carries') **one solid bundle into the next register column.**

LESSON 2: BUNDLING 10 ONES INTO 1 TEN-ROD ↔ [g][÷R] KEY

STEP A: HANDS-ON WITH TEN-RODS & ONES — 1:1 CAD SCENE



10 Unit Counters 10 × 7mm = 70.0 mm | 1 Purple '10' Rod 70.0 mm Solid Tower | Place Value Trade 15 + 10 = 1 R 5

▲ Ten stacked unit counters (10 × 7.0 mm) stand at the exact same 70.0 mm height as one solid Purple '10' rod!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

PLACE VALUE REGROUPING: 15 = 1 TEN + 5 ONES

15 + 10 = 1 R 5

X = 1 Purple '10' | Y = 5 Unit Ones

10×7mm=70mm | 5×7mm=35mm | [g][÷R]

15 [ENTER] 10 [g][÷R] → X=1, Y=5

PRESS BOX ICON & RPN KEYS (SC6):

1. DIV BY 10: 15 ENTER 10 g ÷R

2. TENS/ONES: X=1 Ten | Y=5 Ones

✓ DISPLAY (X) = 1

KEYSTROKES: 15 [ENTER] 10 [g] [÷R]

Dividing 15 by 10 with [g][÷R] puts Quotient = 1 Ten-Rod in X and Remainder = 5 Ones in Y!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 5 | X: 1 → X = 1

THE MAKE-A-TEN BRIDGE • HOW MENTAL ARITHMETIC ACTUALLY WORKS

STEP 3: FLIP SHEET OVER ↻ PAGE 05

05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

5-Anchor Subitizing: 8 = 5 + 3 & 10-Complement + 3 Challenges

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

3 5-ANCHOR SUBITIZING: SEE 8 AS 5 + 3 & 2 SHY OF 10

THE GOAL: Subitize 8 in 0.1 seconds and spot its 10-complement instantly!

1. FILL TOP ROW (5 COUNTERS):

Place 5 red counters across the top row to anchor the benchmark 5.

2. ADD 3 ON BOTTOM ROW:

Place 3 red counters on the bottom row: 5 on top + 3 below = 8 counters total.

3. SPOT 2 EMPTY HOLES:

Notice 2 empty spots remaining in the frame: 8 is exactly 2 shy of 10!

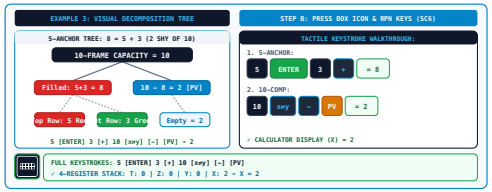
✓ WHY 5-ANCHORS SUPERCHARGE MENTAL ARITHMETIC

The human eye can only subitize up to 4 items at once without counting 1-by-1.
Grouping in 5s lets you recognize 8 as (5 + 3) instantly—no counting needed!
Because a full frame is 10, the two empty holes immediately show that 10 − 8 = 2.
Verify on calculator: 5 [ENTER] 3 [+] (→ 8) | 10 [xʏ] [−] [PV] (→ 2) confirms both!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE



STEP B: PRESS BOX ICON & RPN KEYS (SC6)

FULL KEYSTROKES: 5 [ENTER] 3 [+] 10 [xʏ] [−] [PV]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 2 → X = 2

3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 — **Place 6 red counters on the frame. How many empty spots to make 10?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 — **Use two 10-frames to solve 9 + 6 by moving 1 counter across!**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 — **Build 37 using purple '10' rods and unit counters. How many pieces total?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: Three '10' rods + 7 counters
#2: 4 spots (6 + 4 = 10) | #2: (6 + 1) + 5 = 10 + 5 = 15

FOR PARENTS & TEACHERS (Grades K–3 (Counting, Subitizing, Place Value & Regrouping))

- CCSS.MATH.CONTENT.K.OA.A.4: For any number from 1 to 9, find the number that makes 10.
- CCSS.MATH.CONTENT.1.OA.C.6: Add and subtract within 20 using the making-ten strategy.
- CCSS.MATH.CONTENT.1.NBT.B.2: Understand that the two digits of a two-digit number represent tens and ones.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([▦] Ten-Frame & [PV]) and keystrokes on your calculator:
   • **5 [ENTER] 3 [+] 10 [xʏ] [−] [PV] → X = 2**

CHECK OFF YOUR MASTERY MILESTONES:

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:



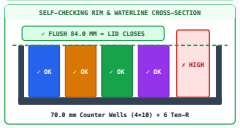
✓ FOUR 10-COUNTER TOWERS (70.0 MM LEVEL CHECK)

The box has 4 round wells that hold exactly 10 stacked counters each (4 × 10 = 40 counters), plus a side bay for the 6 purple '10' rods and 6 gold operator blocks (+, −, =).
When all 40 counters are stacked in the 4 wells, all four towers stand at the exact same **70.0** mm height—so you can spot a missing counter from across the room!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE



▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

DUAL 10-FRAME STAGE LID

- The box lid has two built-in 2×5 ten-frames (20 spots total) for hands-on addition.
- Each counter is red on one side and green on the other—just flip to change signs or addends!
- Easy care: rinse in lukewarm soapy water and air dry.

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ BASE-10 4-POINT CLEAN-UP CHECKLIST

1. FOUR 70MM WELLS:	Stack 10 counters in each of the 4 round wells (all level at 70.0 mm).
2. SEAT 6 TEN-RODS:	Place all 6 Purple '10' rods and 6 Gold sign blocks into the side bay.
3. CLEAR 10-FRAME LID:	Remove any counters still seated in the 20 top lid sockets.
4. FOLD CARD & SNAP:	Lay this folded guide inside and snap the 10-Frame Lid shut!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

FOUR 10-COUNTER WELLS • EQUAL 70MM STACKS PROVE NOTHING IS LOST

NEXT: CONNECTED LAB ▾ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 ◂ YOUR BOX **Ten-Frames Lab** Early Counting & Regrouping	**DOMI-01** **Math Domino Studio** Doubles & 12-Frame Part-Whole	**FRAC-01** **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	**ALGB-01** **Algebra Studio** Quadratic Area Arrays & Factoring	**TIME-01** **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	**VOLV-01** **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	**SC-32-HW** **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

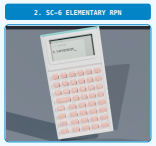
WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. BASE-10 BOX LID ICON



2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN



▲ **BOX ICON → CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- 10 stacked counters (70.0mm) match the height of ten [2] Prime Blocks or ten 1/12th Fraction Blocks.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Bundling 10 ones into 1 ten-rod builds the intuition behind every base-10 calculator register.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 15 (1 Ten + 5 Ones) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [▦] Ten-Frame & [PV]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [▦] Ten-Frame & [PV] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 8 [ENTER] 2 [+] 5 [+] [▦] → X = 15
• Lesson 2: 15 [ENTER] 10 [g] [÷R] → X = 1

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 54-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

PART ID REFERENCE TABLE (54 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

PART NAME	QTY	COLOR / FINISH	PART ID
2-Color Flip Counters	40x	Red / Green	BASE-TENF-ad3
Solid '10' Regrouping Rods	6x	Purple	BASE-TENF-fec
Math Sign Blocks (+ − =)	6x	Gold	BASE-TENF-fec
3x1 Case & 10-Frame Lid	2x	Black	BASE-TEN-090

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.

Enter any Part ID above at **stackcalc.io/parts** to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

Red	Green	Purple	Gold
Counter Side A (Part 1)	Counter Side B (Part 2)	Solid '10' Rod (6x)	Operator Blocks (+ − =)

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU BASE-10-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/ten-frames/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 1 (Ten-Frames & Place Value) • SC-32 Manual Ch. 1
STANDARDS: CCSS K.OA.A.4, 1.OA.C.6, 1.NBT.B.2 • TEKS K.2I, 1.3D (Subitizing & Base-10)
WATERLINE SPEC: 70.0 mm Counter Wells (4×10) + 6 Ten-Rod Bays

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):
Ex 1: 8 [ENTER] 2 [+] 5 [+] [▦] → X=15
Ex 2: 15 [ENTER] 10 [g] [÷R] → X=1
Ex 3: 5 [ENTER] 3 [+] 10 [xʏ] [−] [PV] → X=2

BOX ICON KEY: [▦] Ten-Frame & [PV]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!